

PAVAN KUMAR KETHA (He/Him)

☎ +91 9704710470

★ Visakhapatnam, Andhra Pradesh

✉ k.pavan.5905@gmail.com

🌐 www.pavankumarketha.me

CAREER OBJECTIVE

Motivated developer seeking an opportunity to contribute to a dynamic team, building efficient applications and grow professionally in a collaborative environment.

EDUCATION

ABC ID - 589857167785

B. TECH – Computer Science and Engineering
Dadi Institute of Engineering & Technology | Anakapalle

Pursuing
Percentage – 72.3%
(Till – III-II)

Intermediate – MPC – TS Board
Vishwa Chaitanya Junior College | Hyderabad

2021-2023
Percentage – 83.9%

SSC – AP Board
Bhashyam Educational Institute | Visakhapatnam

2021
Percentage – 100 %

TECHNICAL SKILLS

- **Programming Languages:** Python, Java
- **AI Engineering:** RAG, MCP, LangChain
- **Tools & Platform:** GitHub, Docker, Vercel, Railway, Codex, Cursor
- **Database:** SQL, Azure SQL, Azure Blob Storage

PROJECTS

Disaster Insight Hub Assistant

Description: Flask-based Retrieval-Augmented Generation (RAG) experience that classifies crisis-related text, fetches safety playbooks, and drafts natural responses through Ollama Cloud (gpt-oss:120b-cloud). The Random Forest classifier (multi-output) still handles crisis detection, while TF-IDF retrieval grounds the assistant in your curated Excel knowledge base.

Tech Stack: Python, React, Tailwind CSS, FastAPI, MongoDB, Ollama, LangChain

Repository Link: <https://github.com/Pavan-5905/Disaster-Insight-Hub-Assistant>

Pothole Detection Assistant

Description: Developed an AI-powered pothole detection system that identifies potholes from road images or video streams using computer vision and deep learning techniques. The application provides real-time detection with an intuitive user interface, enabling efficient road monitoring and assisting authorities in prioritizing maintenance to improve road safety.

Tech Stack: Python, YOLO, OpenCV, NumPy

Repository Link: <https://github.com/Pavan-5905/Pothole-Detection-Assistant>

Smart Weather Monitoring System (MINI PROJECT)

Description: Developed an offline smart weather monitoring system using Arduino ESP8266 and environmental sensors to provide real-time temperature, humidity, and rainfall data for rural farmers without internet connectivity.

Tech Stack: Arduino ESP8266, Embedded C, DHT11 Sensor, Rain Sensor, LDR, LCD Display (I2C)

Repository Link: <https://github.com/Pavan-5905/Smart-Weather-Monitoring-System>

CERTIFICATIONS

Python – <i>Elite+Silver</i>	NPTEL	Nov, 2025
Front – End Web Development	eDX / IBM	Jul, 2024
Adobe Creative Cloud	Udemy	Jan, 2024
MS Office - .ppt, .docx, .xls	Corporate Finance Institute(CFI) / Udemy	Jul, 2021

WORKSHOPS & INDUSTRIAL VISITS

Git & GitHub – Jul, 2026 (Trainer)

Key Learnings:

- Basics of Git
- Git Architecture
- Remote Repository Hosting Services
- Working with branches

Visit To Naval Dockyard – Jun, 2026

Key Learnings:

- What is Hydrography
- Importance of Hydrography
- Mapping of Coastal Areas
- Inside a Survey and War Ships

Quantum Computing – Feb, 2026

Key Learnings:

- What is Quantum Computing?
- Basics of Quantum Computing
- Research in Quantum Computing

Parallel Programming Workshop – Feb, 2024

Key Learnings:

- Parallel Programming
- Super Computing

ACHIEVEMENTS & LEADERSHIP

- Secured NPTEL Elite + Silver in Python in November, 2026
- **STUDENT CHAIR** of the CSI Student Branch – DIET from 2025 - 2026
- **DVC** (Diet Volunteering Club) for volunteering services in the campus level programs
- Volunteered at CII – **Partnership Summit** held at Visakhapatnam in April, 2025

COMMUNITY SERVICES

Organic Farming – April & May, 2025

Conducted an 8-week organic farming awareness camp, teaching sustainable and chemical-free practices.

- Bio-Fertilizers
- Vermi Compost
- Natural Manures
- Crop Rotation Techniques
- Water harvesting Methods

SOCIAL LINKS

www.linkedin.com/in/pavan-kumar-ketha

<https://github.com/Pavan-5905>

https://www.hackerrank.com/k_pavan_5905

https://twitter.com/pavan_917

<https://www.instagram.com/thepavankumar.k>

-LinkedIn

-GitHub

-HackerRank

-X(Twitter)

-Instagram

Date: 16th June, 2026

Place: Visakhapatnam, Andhra Pradesh

Signature
(Pavan Kumar Ketha)